

Feature Selection Analysis

Student Exam Performance Dataset

630,000

Dataset Rows

13

Columns

9

Methods Applied

26%

Features Reduced

Project Overview

Problem Statement

Given a dataset of 630,000 students (13 columns), identify which features truly drive exam_score — and prove it mathematically — while discarding redundant noise.

STEP 01

Baseline Model

All 23 features
R² benchmark set

STEP 02

RFECV

Optimal features
scientifically found

STEP 03

9 Methods

Filter · Wrapper
Embedded validated

STEP 04

Final Model

17 features
Same R² · Less noise

Dataset & Data Quality

Dataset Features (13 columns)

Feature	Type	Role
age	Numeric	Input
study_hours	Numeric	★ Key Input
class_attendance	Numeric	Input
sleep_hours	Numeric	Input
gender / course	Categorical	Input
sleep_quality	Categorical	Input
study_method	Categorical	Input
facility_rating	Categorical	Input
exam_difficulty	Categorical	Input
internet_access	Categorical	Input
exam_score	Numeric	🎯 Target

Data Quality Check

- ✓ Missing Values **None found**
- ✓ Duplicate Rows **None found**
- ✓ Outliers (Boxplot) **None detected**

Preprocessing Steps

- Dropped id column (non-predictive)
- One-hot encoded 7 categorical columns
- drop_first=True to avoid multicollinearity
- Result: 13 raw → 23 encoded features
- 80/20 Train-Test split (random_state=42)

Baseline Model — Before Feature Selection

Why build a baseline first? → We need a benchmark R^2 to prove feature selection worked. Without this, we have nothing to compare against.

Code Snippet

```
X_train, X_test, y_train, y_test =  
    train_test_split(X, y, test_size=0.2,  
                    random_state=42)  
  
model_baseline = LinearRegression()  
model_baseline.fit(X_train, y_train)  
  
r2 = r2_score(y_test, y_pred)  
rmse = sqrt(mean_squared_error(...))
```

Train $R^2 \approx$ Test $R^2 \rightarrow$ Difference $< 0.05 \rightarrow$ Model generalizes well to unseen data

0.7782

Training R^2

0.7780

Test R^2

~8.5

Train RMSE

~8.5

Test RMSE

✓ Good Fit — No Overfitting Detected

RFECV — Finding Optimal Features Scientifically

1

Start with
23 features

2

Remove weakest
feature

3

5-Fold CV
 R^2 score

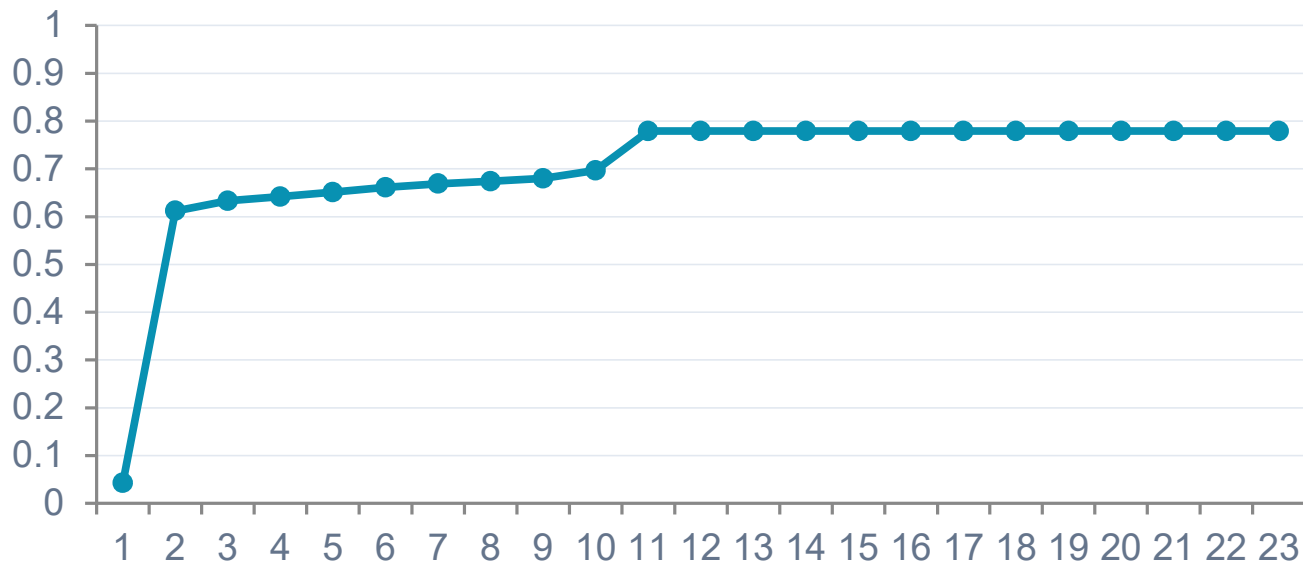
4

Repeat until
1 feature

5

Pick peak
 R^2 count

RFECV: R^2 vs Number of Features



Learning Zone

1 → 10 · R^2 grows rapidly

Saturation Zone

11 → 17 · Only 0.0128% gain

Noise Zone

18 → 23 · R^2 drops — noise added

RFECV Selected

17 features · R^2 peak = 0.7791

💡 Production tip: 11 features = same R^2 (0.0128% diff)

9 Feature Selection Methods Applied

FILTER METHODS	WRAPPER METHODS	EMBEDDED METHODS
1. Correlation	1. RFECV	1. Lasso Regularization
2. Variance Threshold	2. Recursive Feature Elimination	2. Random Forest Importance
3. Chi-Square	3. Sequential Feature Selection	
4. Mutual Information		
5. VIF		
All methods use k = 17 (determined by RFECV) · No guessing · Scientifically validated		

Filter Methods — Key Results

1. Correlation

Measures linear relationship with exam_score

Top Features: study_hours ($r=0.762$), class_attendance ($r=0.36$), sleep_quality_good, sleep_hours

💡 study_hours has strongest linear signal

study_hours ✓

2. Variance Threshold

Removes near-zero variance features (threshold=0.01)

Top Features: 22/23 features retained — most have sufficient variability

💡 Dataset is diverse; no near-constant features

study_hours ✓

3. Chi-Square

Statistical dependency of categorical features on target

Top Features: study_hours, class_attendance, sleep_hours, sleep_quality, study_method, facility_rating

💡 Confirms study & sleep habits dominate

study_hours ✓

4. Mutual Information

Non-linear information each feature provides about exam_score

Top Features: study_hours, class_attendance, sleep_hours, sleep_quality_good, sleep_quality_poor

💡 Consistent with correlation — linear relationships hold

study_hours ✓

5. VIF (Multicollinearity)

Detects highly correlated features (VIF > 5 flagged)

Top Features: 17 features with acceptable VIF retained

💡 age, class_attendance, sleep_hours show high VIF

study_hours ✓

Wrapper & Embedded Methods

WRAPPER METHODS

6. RFE (n=17)

Recursively removes weakest features based on model coefficients

Selected: study_hours, class_attendance, sleep_hours, sleep_quality, study_method, facility_rating + 11 more

✓ Model-based ranking

⚠ Manual n required

7. SFS (n=17)

Adds features step-by-step, evaluating combinations

Selected: study_hours, class_attendance, sleep_hours, sleep_quality, study_method, facility_rating + 11 more

✓ Evaluates combinations

⚠ Very slow

EMBEDDED METHODS

8. Lasso ($\alpha=0.1$)

Shrinks weak feature coefficients to zero automatically

Selected: study_method_self-study, facility_rating_low, study_method_online videos, study_hours, study_method_group study, sleep_quality_poor

💡 Highlighted study_method family — unique finding

9. Random Forest

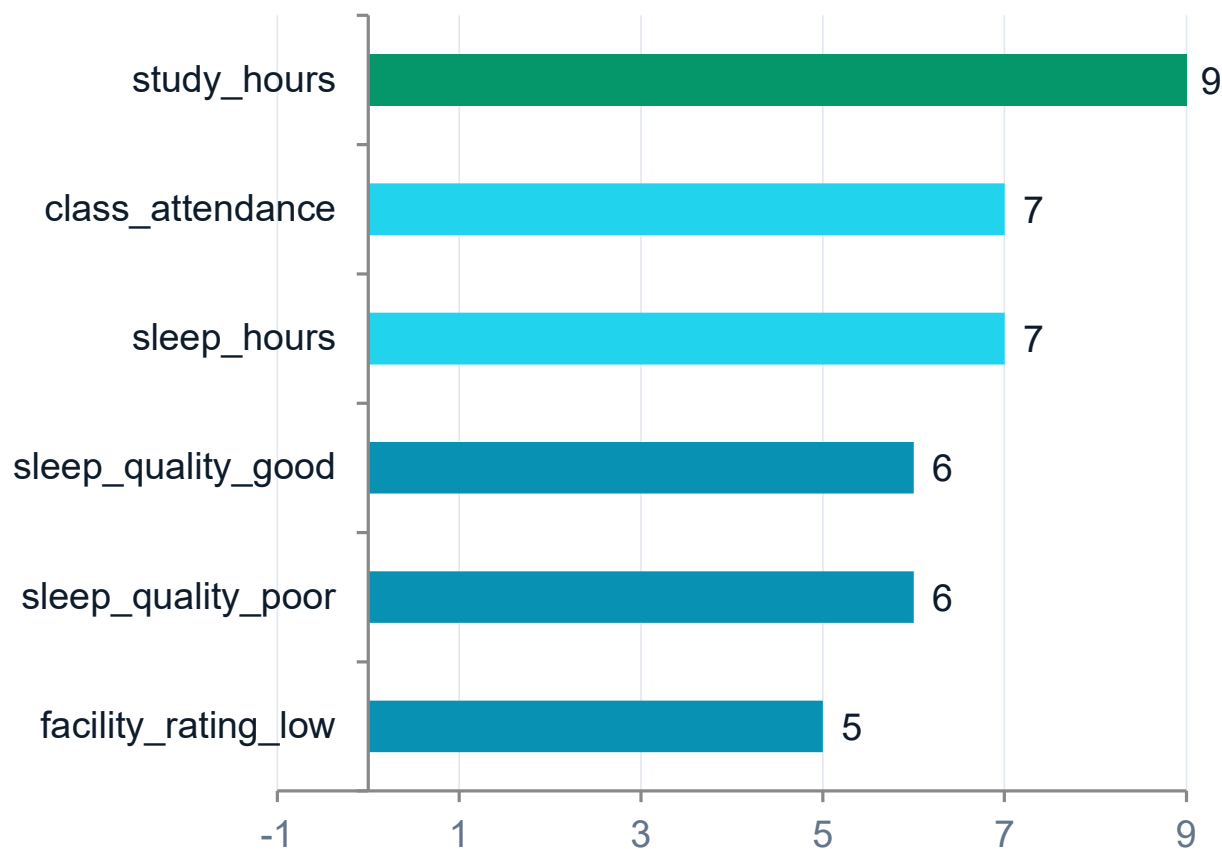
Importance via impurity reduction across 100 decision trees

Selected: study_hours (#1), class_attendance (#2), sleep_hours (#3)

💡 Captures non-linear interactions — most reliable embedded method

Feature Consistency Across All 9 Methods

Feature Selected By N Methods (out of 9)



study_hours

Selected by EVERY method — undisputed #1 predictor

9/9

class_attendance

Strong predictor — attendance drives performance

7/9

sleep_hours

Sleep duration matters — supports well-being research

7/9

sleep_quality

Quality matters as much as quantity

6/9

facility_rating_low

Learning environment has moderate impact

5/9

Before vs After Feature Selection

Metric	Baseline (23 features)	Final (17 features)	Change
Features Used	23	17	↓ 26% fewer
Test R ²	0.7780	0.7781	✓ Same
Train R ²	0.7782	0.7782	✓ Same
Test RMSE	~8.5	~8.5	✓ Same
Overfitting?	No	No	✓ Clean
Model Complexity	High	Low	✓ Simpler

0

Performance Lost

6

Redundant Features
Removed

26% Complexity Reduced

Key Takeaways

study_hours is King

Selected by ALL 9 methods. Strongest predictor of exam performance with $r=0.762$.

RFECV = Best Method

Proved 17 features optimal via 5-fold CV. Automated — no guessing, no bias.

6 Features Were Pure Noise

Removing them caused ZERO performance loss. Same $R^2=0.778$ with 26% less complexity.

Production Ready Insight

11 features achieve same R^2 (0.0128% diff). Reduces data collection cost in deployment.

No Overfitting Throughout

Train $\approx$ Test R^2 in all models. Generalizes well to unseen student data.

Comprehensive Validation

9 methods $\times$ 3 categories $\times$ RFECV confirmation = robust, reproducible conclusion.